Online Supplementary Material

Vilimir Yordanov*

Appendix F to "On iterated Lorenz curves with applications: the multivariate case"

Abstract: This file contains Appendix F, which is provided as supplementary material.

Keywords: Lorenz curve, iteration, contraction mapping, golden section, copula

MSC 2020: 39B12, 60F99, 62G30, 62H05, 62P05

***Corresponding author: Vilimir Yordanov,** Technical University Vienna, Financial and Actuarial Mathematics; Vienna University of Economics and Business, Vienna Graduate School of Finance, Vienna, Austria, E-mail: vilimir.yordanov@tuwien.ac.at, vilimir.yordanov@vgsf.ac.at; non-academic e-mail: villyjord@gmail.com

Appendix F

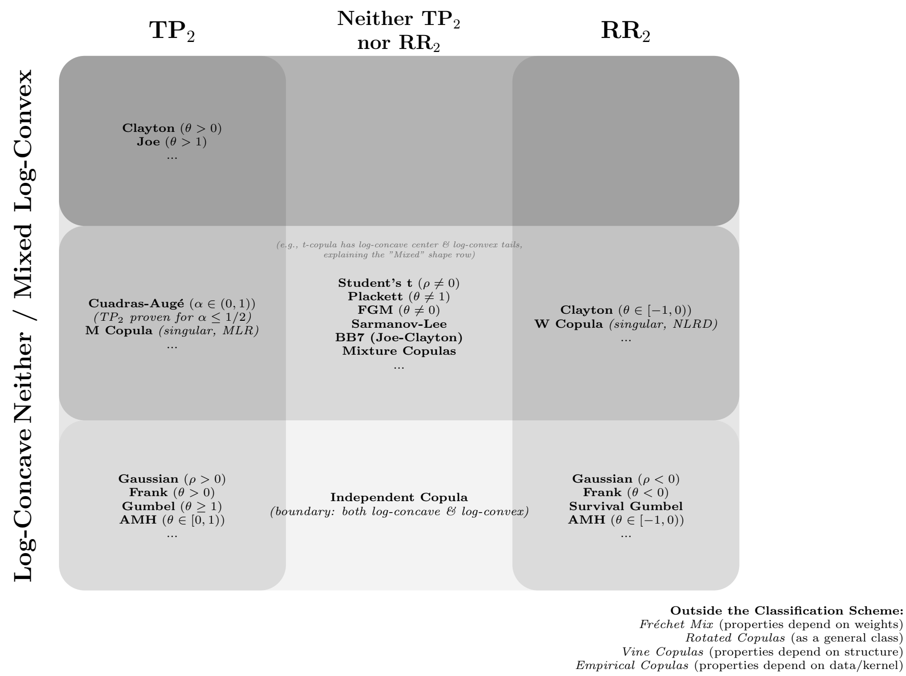

Figure F1: Copulas properties

References

[1] Aaberge, R. (2000). Characterizations of Lorenz curves and income distributions. Social Choice and Welfare, 17, 639-653.

[2] Arnold, B. C. (1983). Pareto Distributions. International Cooperative Publishing House. Fairland, Maryland.

[3] Arnold, B. C. (1987). Majorization and the Lorenz Order: A Brief Introduction. Lecture Notes in Statistics.

[4] Arnold, B. C. (2015). Pareto Distributions, 2nd Ed, Chapman and Hall/CRC.

[5] Billingsley, P. (1995). Probability and measure. John Wiley & Sons.

[6] Brown, L. D., Johnstone, I. M., & MacGibbon, K. B. (1981). Variation diminishing transformations: a direct approach to total positivity and its statistical applications. Journal of the American Statistical Association, 76(376), 824-832.

[7] Cambanis, S., Simons, G., & Stout, W. (1976). Inequalities for E k (x, y) when the marginals are fixed. Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete, 36(4), 285-294.

[8] Cheng, S. S., & Li, W. (2008). Analytic solutions of functional equations. World Scientific.

[9] Colangelo, A., Scarsini, M., & Shaked, M. (2006). Some positive dependence stochastic orders. Journal of Multivariate Analysis, 97(1), 46-78.

[10] Dall'Aglio, G. (1972). Fréchet classes and compatibility of distribution functions. In Symposia mathematica (Vol. 9, pp. 131-150).

[11] Denuit, M., Dhaene, J., Goovaerts, M., & Kaas, R. (2006). Actuarial theory for dependent risks: measures, orders and models. John Wiley & Sons.

[12] Devaney, R. L. (2021). An Introduction To Chaotic Dynamical Systems.

[13] Efron, B. (1965). Increasing properties of Polya frequency function. The Annals of Mathematical Statistics, 272-279.

[14] Evans, L. C. (2022). Partial differential equations (Vol. 19). American Mathematical Society.

[15] Fallat, S., Lauritzen, S., Sadeghi, K., Uhler, C., Wermuth, N., & Zwiernik, P. (2017). Total positivity in Markov structures. The Annals of Statistics, 1152-1184.

[16] Fejer, L. (1906). Über die Fourierreihen, II. Math. Naturwiss. Anz Ungar. Akad. Wiss, 24(369.390) (In German).

[17] Filipów, R. (2013). On Hindman spaces and the Bolzano–Weierstrass property. Topology and its Applications, 160(15), 2003-2011.

[18] Filipów, R., Kowitz, K., & Kwela, A. (2024). A unified approach to Hindman, Ramsey, and van der Waerden spaces. The Journal of Symbolic Logic, 1-53.

[19] Folland, G. B. (1999). Real analysis: modern techniques and their applications. John Wiley & Sons.

[20] Furstenberg, H. (2014). Recurrence in ergodic theory and combinatorial number theory. Princeton University Press.

[21] Gill, J. (2012). Convergence of infinite compositions of complex functions. Comm. Anal. Theory Contin. Fractions, 19, 1-27.

[22] Gill, J. (2017). A Primer on the Elementary Theory of Infinite Compositions of Complex Functions. Communications in the Analytic Theory of Continued Fractions. XXIII.

[23] Glasserman, P., & Pirjol, D. (2024). When Are Option Prices TP2?. Available at SSRN 4887499.

[24] Grafakos, L. (2024). Fundamentals of Fourier Analysis. Springer.

[25] Halperin, I., & Schwartz, L. (1952). Introduction to the Theory of Distributions. University of Toronto Press.

[26] Hardy, G. H., Littlewood, J. E., & Pólya, G. (1952). Inequalities. Cambridge University Press.

[27] Heil, C. (2009). Approximate Identities. Lecture notes, available online, https://heil.math.gatech.edu/7338/fall09/approxid.pdf.

[28] Hindman, N., & Strauss, D. (1998). Algebra in the Stone-Čech compactification: theory and applications (Vol. 27). Walter de Gruyter.

[29] Hörmander, L. (1990). The Analysis of Linear Partial Differential Operators I. Grundlehren der mathematischen Wissenschaften, 256, 2nd Ed, Springer-Verlag.

[30] Ibrahim, I. M. (2016). Using New Criteria to Compare between Some Robust Method and Ordinary Least Squares in Multiple Regression with Application on Wheat Data in Iraq, Faculty of Economics and Business Administration, Sofia University.

[31] Ignatov, Z., & Yordanov, V. (2025). On Iterated Lorenz curves. Annual of Sofia University "St.Kliment Ohridski", vol. 24, 71 –118.

[32] Iordanov, I., & Chervenov, N. (2016). Copulas on Sobolev Spaces. Serdica Math J, 42(3-4), 335-360.

[33] Joe, H. (1997). Multivariate models and multivariate dependence concepts. CRC press.

[34] Kallenberg, O. (2021). Foundations of modern probability. New York, NY: Springer New York.

[35] Karlin, S. (1968). Total Positivity, Stanford, CA, Stanford University Press.

[36] Karlin, S., & Rinott, Y. (1980). Classes of orderings of measures and related correlation inequalities. I. Multivariate totally positive distributions. Journal of Multivariate Analysis, 10(4), 467-498.

[37] Karlin, S., & Rinott, Y. (1980). Classes of orderings of measures and related correlation inequalities II. Multivariate reverse rule distributions. Journal of Multivariate Analysis, 10(4), 499-516.

[38] Kämpke, T., & Radermacher, F. J. (2015). Income modeling and balancing. Lecture Notes in Economics and Mathematical Systems.

[39] Keen, L., & Lakic, N. (2007). Hyperbolic geometry from a local viewpoint (Vol. 68). Cambridge University Press.

[40] Kojman, M. (2002). Hindman spaces. Proceedings of the American Mathematical Society, 130(6), 1597-1602.

[41] Kolmogorov, A. N., & Fomin, S. V. (1975). Introductory real analysis. Courier Corporation.

[42] Koshevoy, G. (1995). Multivariate Lorenz majorization. Social Choice and Welfare, 93-102.

[43] Lehmann, E. L. (1966). Some Concepts of Dependence. The Annals of Mathematical Statistics, 37(5), 1137-1153.

[44] Lehmann, E. L., & Romano, J. P. (2006). Testing Statistical Hypotheses. Springer Science & Business Media.

[45] Marshall, A. W., Olkin, I., Arnold, B. C. (2011). Inequalities: Theory of Majorization and Its Applications. Springer.

[46] Meyer, M., & Strulovici, B. (2013). The supermodular stochastic ordering. CEPR Discussion Paper No. DP9486.

[47] Mitrinovic, D. S., Pecaric, J., & Fink, A. M. (2013). Classical and new inequalities in analysis (Vol. 61). Springer Science & Business Media.

[48] Müller, A., & Stoyan, D. (2002). Comparison methods for queues and other stochastic models. Wiley&sons: New York, NY.

[49] Nelsen, R. B. (2006). An introduction to copulas. Springer.

[50] Pellerey, F., & Navarro, J. (2022). Stochastic monotonicity of dependent variables given their sum. Test, 31(2), 543-561.

[51] Pinkus, A. (2010). Totally positive matrices (No. 181). Cambridge University Press.

[52] Polyanin, P., & Manzhirov, A. V. (2008). Handbook of integral equations. Chapman and Hall/CRC.

[53] Puccetti, G., & Wang, R. (2015). Extremal Dependence Concepts. arXiv preprint arXiv:1512.03232.

[54] Pugh, C. C. (2015). Real Mathematical Analysis. Springer.

[55] Rudin, W. (1987). Real and complex analysis. McGraw-Hill, Inc..

[56] Rüschendorf, L. (1980). Inequalities for the expectation of Δ-monotone functions. Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete, 54(3), 341-349.

[57] Rüschendorf, L. (1983). Solution of a statistical optimization problem by rearrangement methods. Metrika, 30(1), 55-61.

[58] Rüschendorf, L. (2004). Comparison of multivariate risks and positive dependence. Journal of Applied Probability, 41(2), 391-406.

[59] Rüschendorf, L. (2013). Mathematical risk analysis. Springer Ser. Oper. Res. Financ. Eng. Springer, Heidelberg.

[60] Rüschendorf, L. (2017). Improved Hoeffding–Fréchet bounds and applications to VaR estimates. In Copulas and Dependence Models with Applications: Contributions in Honor of Roger B. Nelsen (pp. 181-202). Springer International Publishing.

[61] Rüschendorf, L., Vanduffel, S., & Bernard, C. (2024). Model risk management: risk bounds under uncertainty. Cambridge University Press.

[62] Sarabia, J. M., & Jorda, V. (2020). Lorenz surfaces based on the Sarmanov–Lee distribution with applications to multidimensional inequality in well-being. Mathematics, 8(11), 2095.

[63] Shaked, M., & Shanthikumar, J. G. (Eds.). (2007). Stochastic orders. New York, NY: Springer New York.

[64] Shalit, H., & Yitzhaki, S. (1984). Mean-Gini, portfolio theory, and the pricing of risky assets. The Journal of Finance, 39(5), 1449-1468.

[65] Sklar, A. (1973). Random variables, joint distribution functions, and copulas. Kybernetika, 9(6), 449-460.

[66] Stein, E. M., & Shakarchi, R. (2011). Fourier analysis: an introduction (Vol. 1). Princeton University Press.

[67] Taguchi, T. (1972). On the two-dimensional concentration surface and extensions of concentration coefficient and pareto distribution to the two dimensional case—I: On an application of differential geometric methods to statistical analysis. Annals of the Institute of Statistical Mathematics, 24(1), 355-381.

[68] Taguchi, T. (1972). On the two-dimensional concentration surface and extensions of concentration coefficient and pareto distribution to the two dimensional case—II: On an application of differential geometric methods to statistical analysis. Annals of the Institute of Statistical Mathematics, 24(1), 599-619.

[69] Tchen, A. H. (1980). Inequalities for distributions with given marginals. The Annals of Probability, 814-827.

[70] Teschl, G. (2001). Nonlinear functional analysis. Lecture notes in Math, Vienna Univ., Austria.

[71] Vladimirov, V. S. (1981). Equations of mathematical physics. Moscow Izdatel Nauka (In Russian).

[72] Whitt, W. (1982). Multivariate monotone likelihood ratio and uniform conditional stochastic order. Journal of Applied Probability, 19(3), 695-701.

[73] Zeidler, E. (1986). Nonlinear Functional Analysis and its Applications. Fixed-point theorems, Springer.